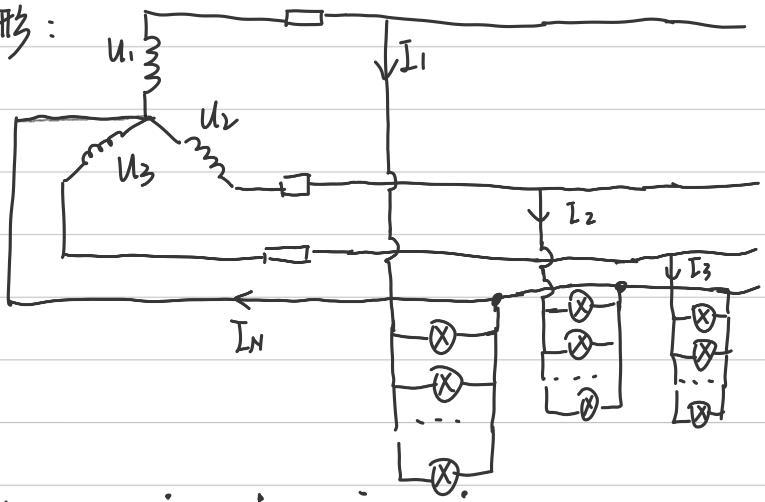
2.4.2.

(1) ∵是三相四线制

∴星形：



(2) ∵满载 ∴ $\dot{I}_N = \dot{I}_1 + \dot{I}_2 + \dot{I}_3 = 0$ $\sqrt{3} U \cdot I \cos\varphi = P$

∴ $I = P / (\sqrt{3} U \cdot \cos\varphi) = 10A$.

(3) $I_1' = \frac{1}{2} I_1 = 5A$ $I_2' = \frac{3}{4} I_2 = 7.5A$. $I_3' = I_3 = 10A$.

$\dot{I}_N' = \dot{I}_1' + \dot{I}_2' + \dot{I}_3' = 5\angle 0° + 7.5\angle -120° + 10\angle 120° = 4.33\angle -30°\ A$

2.4.4.

(1) 对三角形连接. $I_\Delta = \frac{\sqrt{3} U_L}{|Z|}$

∴ $I_\Delta / I_Y = 3$.

对星形连接 $I_Y = \frac{U_L}{\sqrt{3}|Z|}$

(2) $P_\Delta = \sqrt{3} U_L I_\Delta \cos\varphi$. $P_Y = \sqrt{3} U_L I_Y \cos\varphi$.

$P_\Delta / P_Y = I_\Delta / I_Y = 3$.

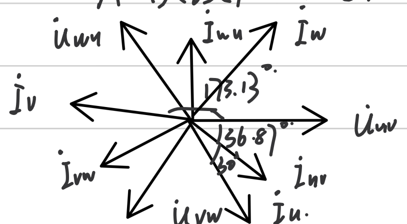
2.4.6

(1) $\dot{I}_{uv} = \frac{\dot{U}_{uv}}{Z} = \frac{380\angle 0°}{32+24j} = 9.5\angle -36.87°$. $\dot{I}_{vw} = 9.5\angle(-36.87° - 120°) = 9.5\angle -156.87°$.

$\dot{I}_{wu} = 9.5\angle(-156.87° - 120°) = 9.5\angle -276.87°$

$\dot{I}_u = \dot{I}_{uv} + \dot{I}_{wu} = 16.45\angle -66.87°\ A$

∴ A_1 读数为 16.45A A_2 读数为 9.5A



$i_u = \sqrt{2} \times 16.45 \sin(314t - 66.87°)\ A$

(2) $\dot{I}_{uv} = 9.5\angle -36.87^\circ A.$

$\dot{I}_{vw} = -\dot{U}_{uv}/(2Z) = 4.75\angle 143.13^\circ A.$

$\dot{I}_u = \dot{I}_{uv} - \dot{I}_{vw} = 14.25\angle -36.87^\circ A.$

$\therefore A_1 : 14.25A \quad A_2 : 4.75A.$

$\dot{I}_u = 14.25 \times \sqrt{2} \cdot \sin(314t - 36.87^\circ)A.$